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# Yichong ZHANG

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Personal website

## Education Background

**Tsinghua University**, Weiyang College, Beijing, China

Aug. 2023 – Jul. 2027 (expected)

Undergraduate in Mathematics and Physics + Electrical Engineering and Automation, GPA: 3.9/4.0, Ranking: 3/52

Awarded to **top 1** students for academic excellence (2023-2024)

Core Courses: *Fundamentals of Software and Hardware Technologies for Artificial Intelligence*, *Equations of Mathematical Physics* *Quantum Mechanics* *Data Structures* *Analog Electronics* *Digital Electronics*

**Harvard College, Harvard University**, Cambridge, MA, United States

Aug. 2025 – May. 2026

Visiting undergraduate student

## Research Interests

My research focuses on building efficient and reliable AI systems through cross-stack optimization and automated systems development. My current interests include:

**Systems and architecture for emerging AI workloads**, particularly efficient real-world deployment of embodied AI.

**Reliable code generation and optimization with coding agents**, with a focus on testing & verification.

## Publications

**Yichong Zhang**, Genghan Zhang, Zishen Wan, Kunle Olukotun, Olivia Hsu, “Behavioral Test Diversity: Quantifying Test Suite Diversity Through Agentic Code Repair,” Submitted to ICLR 2027, *paper*

Jiayi Qian, **Yichong Zhang**, Hanchen Yang, Chun Tao, Souvik Kundu, Zishen Wan, Tushar Krishna, “*Serving Agentic Workflows with a Physical-Plan Compiler and Adaptive Runtime*,” Submitted to ASPLOS 2027, *paper*

Ke Hong, Xiuhong Li, Minxu Liu, Qiuli Mao, Tianqi Wu, Zixiao Huang, Lufang Chen, Zhong Wang, **Yichong Zhang**, Zhenhua Zhu, Guohao Dai, Yu Wang, “*Efficient and Adaptable Overlapping for Computation and Communication via Signaling and Reordering*,” accepted by EuroSys, 2026, *paper*

Tianchen Zhao, Ke Hong, Xinhao Yang, Xuefeng Xiao, Huixia Li, Feng Ling, Ruiqi Xie, Siqi Chen, Hongyu Zhu,

**Yichong Zhang**, Yu Wang, “*PAROAttention: Pattern-Aware ReOrdering for Efficient Sparse and Quantized Attention in Visual Generation Models*,” accepted by NeurIPS, 2025, *paper*

## Research Experience

**Measuring and Improving Alignment of Coding Agents through Tests as Specifications**

May. 2026 – present

Advisor: Profs. Kunle Olukotun & Olivia Hsu, Stanford University

Investigated how effectively tests specify intended program behavior for coding agents, identifying limitations of existing test-suite metrics in measuring the diversity of behavioral feedback.

Developed Behavioral Test Diversity (BTD), a new metric that quantifies the diversity of behavioral specifications encoded by a test suite by measuring whether code repairs made for one test also satisfy other tests.

Applied BTD to guide test-suite pruning and expansion, reducing tests supplied as executable specifications by 61–90% across ProgramBench, vLLM, and ComputeEval while preserving full-suite recovery, and achieving 3.2× higher behavioral-diversity gain than a flat expansion baseline.

**Benchmarking and Optimizing Robotic Manipulation Workloads.**

Feb. 2026 – present

Advisor: Profs. Zishen Wan & Vijay Janapa Reddi, Harvard University

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Developed lightweight profiling integrations for robotic manipulation benchmarks, enabling measurements of latency, energy consumption, and hardware utilization alongside task success.

Benchmarked manipulation policies (VLA, WAM and Code-As-Policy) across different tasks (LIBERO, RoboTwin, BEHAVIOR) on different hardware (H100, RTX 5090, Jetson Thor/Orin).

Evaluated dynamic voltage and frequency scaling (DVFS) on Jetson platforms and asynchronous inference across models to assess their effects on workload efficiency and performance.

Validated the profiling framework on a real-world Franka robot arm with an RTX 5090 running pi0.5, extending evaluation beyond simulation-based benchmarks and analyzing task-level efficiency in deployment.

### **Efficiency Optimization for Vision-Language-Action Models**

Sep. 2025 – Feb. 2026

*Advisor: Profs. Vijay Janapa Reddi & Zishen Wan, Harvard SEAS, Harvard University*

Benchmarked cache reuse and 4/8 bit quantization on OpenVLA/OpenVLA-oft/CogACT across LIBERO and safety/robustness benchmarks (VLA\_UQ, LIBERO-Plus/Pro), observing no apparent performance degradation.

Conducted extreme low-bit (2–3 bit) quantization on OpenVLA/OpenVLA-OFT LLM backbone; observed minimal performance drop at 3-bit (–0.9%) and significant degradation at 2-bit (–16%) on the LIBERO benchmark.

Implemented multiple top-rank pruning paradigms on OpenVLA/OpenVLA-oft LLM backbone (e.g., SVD vs. RRQR) and found SVD-based pruning consistently outperforms alternatives across all pruning rank settings.

Analyzed the interpretability of VLA models by studying dominant singular directions of LLM backbone weights, showing that key singular subspaces correspond to disentangled and semantically meaningful action primitives.

### **GPU Systems and Kernel Optimization**

Jan. 2025 – Aug. 2025

*Advisor: Prof. Yu Wang, Department of Electronic Engineering, THU*

Implemented fused CUDA reorder kernels for LayerNorm and RoPE in a hardware–software co-designed sparse-attention system for Video Transformer, achieving ~20× speedup over PyTorch implementations.

Built end-to-end benchmarking pipelines and test benches for PARO-Attention, and conducted kernel-level performance comparisons against FlashAttention2, XAttention and SVG.

Contributed to a signaling-based computation–communication overlap mechanism for GPU workloads and built benchmarking pipelines across GEMM shapes and collective communication primitives, achieving up to 1.65× end-to-end speedup on RTX 4090 GPUs.

### **Extracurricular Activities**

**"A GreatPerhaps" Career Talk Project** Beijing, China

Mar. 2024 – Jan. 2025

*Founder of the project*

Founded and led a cross-university team spanning Tsinghua, Peking, Shanghai Jiao Tong, Fudan, and the University of Washington, producing career-focused interviews and content with ~100,000 cumulative views.

Conducted one-on-one interviews with senior technology executives, including CTOs from Megvii Technology, Purdue Technology, and TeleAI.

Project website: <https://grephap.org>

**Tsinghua University BID (Business Innovation Design) Program** Beijing, China

Sep. 2024 – Dec. 2024

*Member of the program*

Analyzed China's skincare and cosmetics market, emphasizing Kans' livestreaming model, and delivered a presentation which received the “Best Oral Presentation” Award.

### **Skills**

**Language:** Chinese (Native); English (Fluent, with a TOEFL score of 106)

**Programming and Software:** Python, CUDA, C++, Bash, Git, LATEX

**Conference review:** ICLR (2027), NeurIPS 2026 Workshop (Verify-Agents Workshop)